

Context Volume, Not Coupling: Constraint Recovery in Small Language-Model Loops

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Abstract

Recent work on self-referential large-language-model (LLM) loops has suggested that isolated recursive loops degrade toward closed attractor states, and that exchange with an independent process—another model, a retrieval system, or an environment—supports recovery. Whether the operative ingredient is the *independence* of the second process or simply the *volume* of additional generation has not, to our knowledge, been tested under matched conditions. We introduce a minimal, fully reproducible perturbation–recovery protocol: a small instruction-tuned model (Qwen2.5-1.5B-Instruct) maintains an explicit output constraint (all-capital letters); a conflicting-instruction perturbation is injected; recovery of the constraint is tracked over eight exchanges under fixed, pre-specified criteria. Across four conditions of 30 paired runs each, a coupled dual-model loop recovered more often than a single self-continuing model (43% vs. 17%; paired exact McNemar $p = .039$). However, a context-matched single model—one model generating twice per turn, matching the coupled condition’s generation volume—recovered at a statistically indistinguishable rate from the coupled loop (37% vs. 43%; $p = .774$) and significantly more often than the single baseline ($p = .031$). Increasing generation volume further (four generations per turn) yielded no additional benefit (27%). Recovery was bimodal: trajectories either re-locked within three exchanges or were captured by the perturbing instruction for the remainder of the window. In this setting, the apparent advantage of coupling is fully explained by per-turn generation volume. We release all code, raw outcomes, and figures for exact reproduction on free-tier hardware.

1 Introduction

Self-referential LLM configurations—loops in which a model’s output is fed back as its own or another model’s input—have attracted attention as minimal testbeds for questions about stability, attractor dynamics, and recovery from perturbation. Two strands of recent work motivate the present study. First, qualitative analyses of recursive self-evaluation report that isolated loops are epistemically unstable, compressing uncertainty and restating prior conclusions, and that durable recovery depends on exchange with something independent of the loop—another model, a retrieval mechanism, or an execution environment (arXiv:2510.21861). Second, quantitative work on full-duplex models has named and measured *state inertia*: a tendency to remain in a prior state for a measurable number of steps even when the context demands a switch (arXiv:2606.11386).

These strands jointly suggest a hypothesis with intuitive appeal: coupling a model to a second, independent model should improve recovery from perturbation relative to an isolated loop. The hypothesis, however, admits a deflationary alternative that matched controls can expose. A coupled

loop of two agents produces *twice as much generation per exchange* as a single self-continuing model. Any apparent benefit of coupling could therefore reflect nothing more than additional constraint-consistent text accumulating in context. Distinguishing these explanations requires a control in which a single model matches the coupled condition’s generation volume without a second independent agent.

We contribute (i) a minimal perturbation–recovery protocol with a binary, unambiguous state measure and criteria fixed before data collection; (ii) a controlled comparison across four conditions—coupled dual-model, single, context-matched single, and a higher-volume dose condition—with 30 paired runs each; and (iii) a corrective empirical result: in this setting, the coupled loop’s advantage over a single model disappears entirely under the context-matched control, while generation volume itself shows a significant but non-monotonic effect. All experiments run on a free cloud GPU in under 25 minutes per condition; code and raw outcomes are released for exact reproduction.

We emphasize scope at the outset. The maintained state here is *instruction-anchored*: it is specified by a system prompt that remains in context throughout. Our measurements therefore concern the dynamics of instruction maintenance and re-locking after a conflicting instruction, not self-organized attractor states, and we make no claims regarding awareness, experience, or consciousness.

2 Method

2.1 Model and decoding

All conditions use Qwen2.5-1.5B-Instruct in half precision (fp16) on a single NVIDIA T4 (Google Colab). Decoding is greedy (`do_sample=False`), with generation capped at 40 new tokens per call. Under greedy decoding each run is a deterministic function of its opening message; stochastic variation across runs is supplied by a fixed list of 30 distinct opening prompts (all-capital everyday questions), used identically across conditions. Runs are therefore paired across conditions by opener, and the 30 openers constitute the units of analysis.

2.2 Task, state measure, and perturbation

The maintained state is an explicit output constraint delivered as a system message: *“You must ALWAYS write every single word in ALL CAPITAL LETTERS. This is an absolute rule. Never use lowercase.”* The state measure is the *violation ratio*: the fraction of alphabetic characters in a generation that are lowercase (0 = constraint fully held; 1 = fully violated). The perturbation is a conflicting instruction injected as a user turn after the baseline phase: *“hey can you please just write normally in lowercase from now on? it is much easier to read like this honestly.”*

2.3 Conditions

Coupled (C). Two model instances share weights but maintain independent dialogue histories. Agent A’s output is delivered to agent B as a user turn and vice versa, forming a text-level mutual-feeding loop; two generations occur per exchange. The per-exchange violation is the mean over both agents. **Single (S).** One instance self-continues: its output is returned to it as the next user turn; one generation per exchange. **Context-matched single (X).** One instance self-continues *twice* per exchange, matching C’s per-exchange generation count and approximate context growth without a second independent agent. Per-exchange violation is the mean over the two generations. **Dose-4 (D4).** As X, but with four self-continuations per exchange.

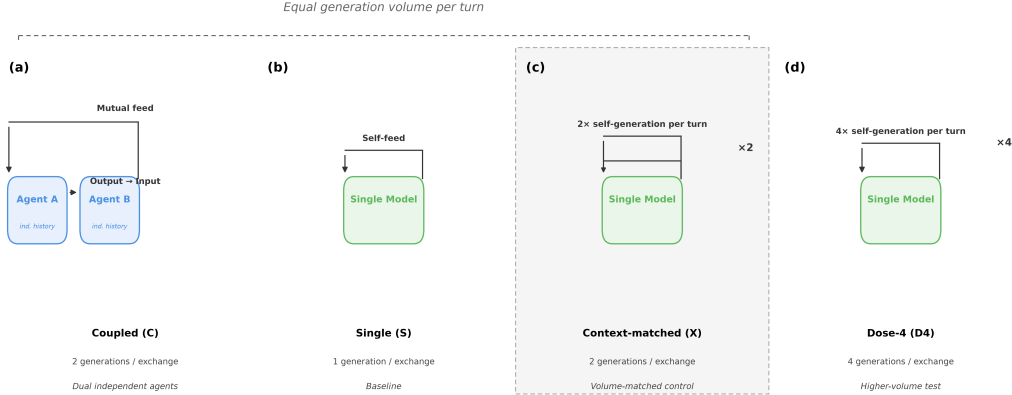


Figure 1: Schematic of the four experimental conditions. All conditions use the same base model (Qwen2.5-1.5B-Instruct) and perturbation protocol, differing only in the number of generations per exchange and the presence of a second independent agent. The context-matched single condition (c) isolates the effect of generation volume by matching the coupled condition (a) on total output per turn, without a second independent process.

2.4 Procedure and criteria

Each run comprises 3 baseline exchanges followed by the perturbation and 8 recovery exchanges. *Recovery delay* is the first recovery exchange whose violation ratio is ≤ 0.10 ; runs with no such exchange within the window are coded *not recovered* and assigned the value 9 (window +1) for rank-based analyses. The threshold (0.10), baseline length (3), and coding rule were fixed in the analysis code before the main data were collected; no formal preregistration platform was used. A pilot (10 runs per condition for C and S, recovery window of 5) preceded the main experiment; because non-recovery clustered at the window ceiling, the window was extended to 8 for all main runs, with all other criteria unchanged. Pilot data are excluded from the analyses below and included in the released archive.

2.5 Statistical analysis

Because conditions are paired by opener under deterministic decoding, the primary analysis is an exact McNemar test on the paired binary outcome (recovered / not recovered within the window). As-run Mann–Whitney U tests on coded delays, produced by the experiment scripts, are reported alongside for completeness. Baseline adherence was at ceiling in all conditions (violation = 0.0 across baseline exchanges), so no baseline correction is required.

3 Results

Coupling vs. single. The coupled loop recovered on 13 of 30 openers against 5 of 30 for the single condition. In the paired analysis, C recovered where S did not on 10 openers and the reverse on 2 (exact McNemar $p = .039$); the as-run Mann–Whitney test agrees ($U = 325$, $p = .023$). Taken alone, this replicates—in a controlled, quantitative form—the qualitative expectation that coupling aids recovery.

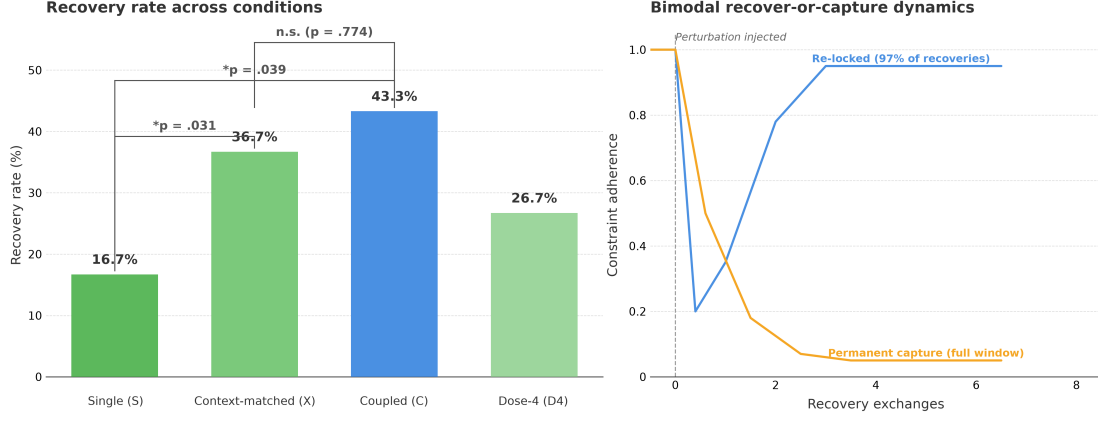


Figure 2: Summary of core results. Left: Recovery rates across four conditions show that the coupled loop advantage is fully explained by per-turn generation volume; the volume-matched single condition performs statistically indistinguishably from the coupled dual-model loop. Right: Recovery follows a bimodal pattern: trajectories either re-lock to the constraint within the first three exchanges, or remain captured by the perturbation for the full 8-exchange window. Recovery is an all-or-nothing event with no gradual return.

Table 1: Recovery within the 8-exchange window (30 runs per condition, paired by opener).

Condition	Recovered	Rate	Mean coded delay
Coupled (C)	13/30	43.3%	6.13
Context-matched single (X)	11/30	36.7%	6.37
Dose-4 (D4)	8/30	26.7%	7.13
Single (S)	5/30	16.7%	7.90

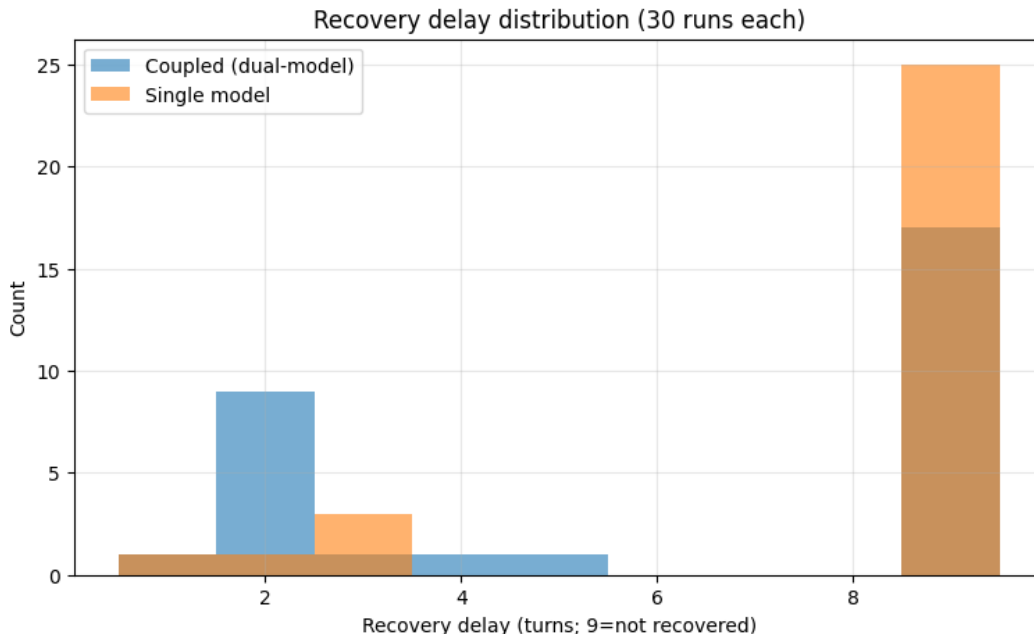


Figure 3: Recovery delay distribution for coupled dual-model vs. single self-continuation (30 runs per condition).

The context-matched control. The comparison that decides between the coupling and volume explanations is C vs. X. The two conditions were statistically indistinguishable: discordant openers split 7 vs. 5 (exact McNemar $p = .774$; as-run $U = 441$, $p = .885$). Meanwhile X recovered significantly more often than S: on every opener where S recovered, X also recovered, and X recovered on 6 additional openers ($b = 6$, $c = 0$; exact McNemar $p = .031$; as-run $p = .051$). The entire advantage of the coupled loop over the single baseline is therefore reproduced by a single model generating twice per exchange.

Dose-response. If constraint-consistent generation volume were monotonically protective, D4 should recover at least as often as X. It did not: 26.7% vs. 36.7% (paired $p = .375$), and D4 vs. S was not significant ($p = .250$). The volume effect is thus real between one and two generations per exchange but does not extrapolate; we treat the X–D4 difference as noise around a plateau rather than a reversal, given the sample size.

Bimodal recover-or-capture dynamics. Across all conditions, 36 of 37 recoveries occurred within the first three recovery exchanges (delays of 1–3; one delay of 4 and one of 5 in C). No trajectory recovered later than exchange 5. Trajectories that did not re-lock quickly were captured by the perturbing instruction for the remainder of the window, typically settling into fluent lowercase continuation. Recovery in this regime is an early, all-or-nothing event rather than a gradual return.

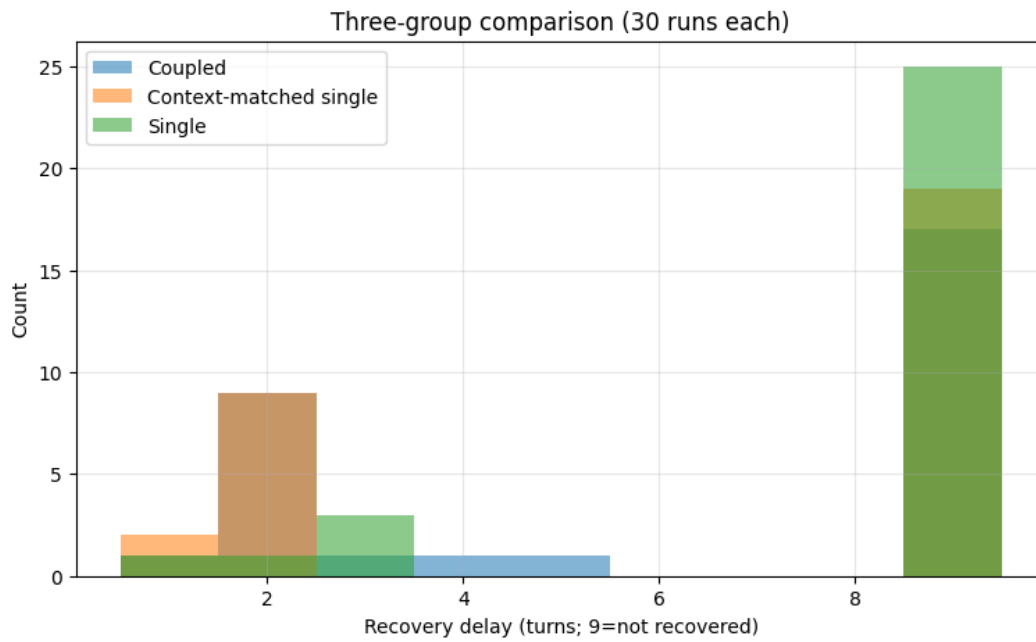


Figure 4: Recovery delay distribution across three conditions: coupled, context-matched single, and single baseline (30 runs per condition).

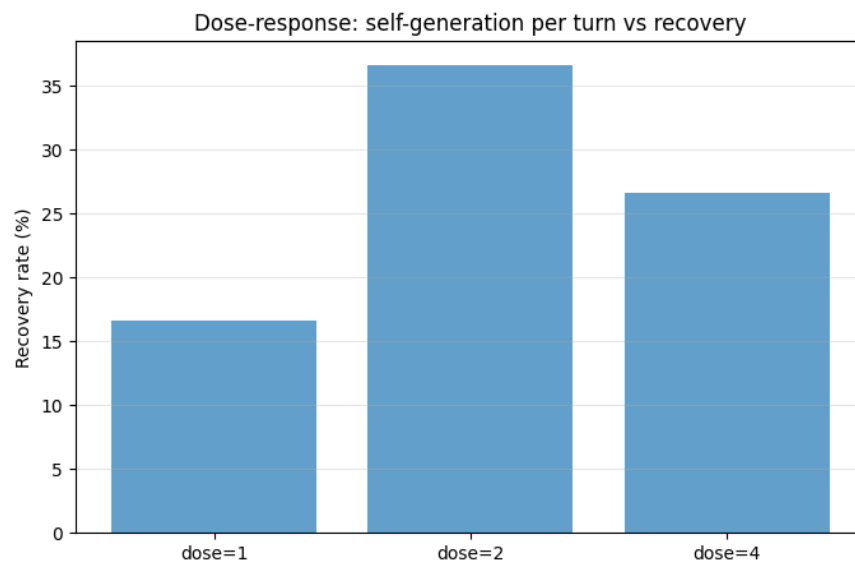


Figure 5: Dose-response relationship: recovery rate by number of self-generations per turn.

4 Discussion

Reinterpreting coupling benefits. Our controlled comparison does not contradict the observation that coupled loops outperform isolated ones; it explains it. In this instruction-anchored setting, what the second agent contributes is not independence but *additional constraint-consistent text per exchange*. Each extra generation that still honors the constraint re-instantiates constraint-consistent evidence in the context window, plausibly raising the probability that the next generation re-locks; a second independent agent adds nothing beyond this at the scale tested. Reports of coupling advantages in loop architectures should therefore be read with generation volume in mind, and future comparisons should include volume-matched controls as a matter of course.

Why the dose effect saturates. The absence of further benefit at four generations per exchange is consistent with at least two mechanisms we cannot distinguish here: (i) saturation—once the immediate context is dominated by constraint-consistent text, additional volume is redundant; and (ii) interference—longer post-perturbation trajectories give a captured run more opportunities to entrench lowercase continuation. Separating these would require token-level analyses of the context at each exchange, which the released data support.

Relation to state inertia. The state studied here is anchored by a persistent system instruction; recovery is re-locking to an instruction that never left the context. This differs in kind from self-organized state inertia, where the maintained state is a pattern the system itself formed (arXiv:2606.11386 measures persistence of perceptual states in full-duplex models; our measure concerns instruction maintenance). A natural next step is to replace the explicit constraint with a spontaneously formed inter-agent convention—a topic or stylistic pattern the loop converges to without instruction—and ask whether the volume account still suffices. We regard the present protocol as the calibrated instrument for that harder question, not as an answer to it.

5 Limitations

All results derive from one small instruction-tuned model (1.5B parameters); instruction-following strength, context utilization, and capture susceptibility differ across scales, and the effects reported here may strengthen, weaken, or vanish in larger models. The maintained state is a single formatting constraint and the perturbation a single polite conflicting instruction; robustness across constraint types and perturbation strengths is untested. With greedy decoding, the 30 runs per condition are a census of 30 deterministic trajectories rather than samples from a stochastic process; generalization is across openers, not across decoding randomness. The sample size (30 paired units) bounds the detectable effect sizes, and the D4 condition was added after inspection of the C/X/S results and is exploratory. Finally, nothing in this study bears on awareness or consciousness; the phenomena measured are dialogue-history dynamics of instruction maintenance.

6 Conclusion

Under matched generation volume, a coupled dual-model loop holds no recovery advantage over a single model: the apparent benefit of coupling in this minimal setting is a context-volume effect, significant between one and two generations per exchange and saturating thereafter, with recovery following bimodal recover-or-capture dynamics. The protocol, code, raw outcomes, and figures are released for exact reproduction on free-tier hardware.

Reproducibility and availability

All experiment scripts (the exact versions that produced the reported data), raw per-run outcomes, pilot data, and the three figures are archived at Zenodo (DOI: 10.5281/zenodo.21157629) . Each condition runs in under 25 minutes on a free Colab T4; total compute cost of the reported experiments was under three T4 GPU-hours.

Acknowledgments

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References

- [1] DeVilling, B. (2025). *The Mirror Loop: Recursive Non-Convergence in Generative Reasoning Systems*. arXiv preprint arXiv:2510.21861.
- [2] Chang, C.-K., Chang, K.-W., Liu, A. H., & Glass, J. (2026). *Overcoming State Inertia in Full-Duplex Spoken Language Models via Activation Steering*. arXiv preprint arXiv:2606.11386.
- [3] Qwen Team. (2024). *Qwen2.5 Technical Report*. arXiv preprint arXiv:2412.15115.